Program : **Diploma in Electronics & Computer Engineering**	
Course Code: **4501**	Course Title: **Fundamentals of Electronic Communication**
Semester: **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester:**60**

Course Objectives:

- To introduce the concept of electronic communication.
- To introduce the principle of amplitude modulation, frequency modulation and pulse modulation.
- To enrich on digital pulse modulation techniques, digital modulation schemes.
- To enrich on the data transmission methods and multiplexing techniques

Course Prerequisites:

Topic	Course code	Course name	Semester
Active and passive components.	2031	Fundamentals of Electrical and Electronics Engineering	2
Power, rms values, peak values.			
Integration, differentiation, trigonometry and mathematical representation of signals.	1002, 2002	Mathematics I & II	1 & 2
Amplifiers, Oscillators and Switching circuits	3043	Electronic Circuits	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain Amplitude Modulation (AM) and Frequency Modulation (FM) techniques.	14	Understanding

CO2	Explain the working of radio transmitters	13	Understanding
CO3	Explain various digital pulse modulation techniques	15	Understanding
CO4	Summarize digital modulation schemes, multiplexing techniques and data transmission methods.	16	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain Amplitude Modulation (AM) and Frequency Modulation (FM) techniques**.		
M1.01	Explain Communication systems.	3	Understanding.
M1.02	Interpret modulation parameters of different AM techniques	5	Understanding.
M1.03	Derive the expression for modulated signals and frequency spectrum of FM	3	Understanding.
M1.04	Solve for modulation index, bandwidth and power for various modulation schemes for given signal parameters	3	Applying

Contents:

Concept of modulation.

Introduction to communication system: Elements of communication systems, Examples of analog communication systems, Frequency bands, Need for modulation.

Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB.

Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM.

Comparison of AM and FM, Applications of AM and FM.

CO2	**Explain the working of radio transmitters and receivers**		
M2.01	Explain the working of AM and FM transmitters	5	Understanding
M2.02	Explain the working of AM and FM receivers	5	Understanding
M2.03	Compare various analog pulse modulation techniques.	3	Understanding.
	Series Test – I	1	

Contents:

AM transmitter: AM Transmitter - low level and high level, AM collector modulator, Balanced modulator

FM transmitter: FM Transmitter(Armstrong method), Pre-Emphasis and De-Emphasis circuits

Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure.

Superheterodyne receiver, AM receiver(block diagram only), choice of IF in receivers , FM receiver(block diagram only), comparison of FM and AM receiver

AM demodulator circuit: diode detector.

Pulse Modulation Schemes: PAM, PWM, PPM, PCM

CO3	**Explain various digital pulse modulation techniques**		
M3.01	Define digital communication systems	1	Remembering
M3.02	Illustrate Sampling and Quantization.	4	Understanding
M3.03	Interpret the PCM systems	4	Understanding
M1.04	Explain DPCM, DM and ADM	6	Understanding

Contents:

Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system,

Sampling and Quantization**:** Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error.

PCM, DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation (Block diagram explanation only)

CO4	**Summarize digital modulation schemes, multiplexing techniques and data transmission methods.**		
M4.01	Summarize digital modulation schemes.	8	Understanding
M4.02	Compare various multiplexing techniques	4	Understanding
M4.03	Describe data transmission methods	4	Understanding
	Series Test – II	1	
Contents: **Digital modulation schemes** Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, **QPSK Multiplexing Techniques:** TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. **Data transmission methods:** Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous			

Text/Reference:

T/R	Book Title/Author
T1	Electronic Communication Systems – George Kennedy – TMH
T2	Communication Systems - Simon Haykin
R1	Communication Systems(Analog and Digital) – Sanjay Sharma
R2	Electronic Communication Systems - Wayne Thomasi.
R3	Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014
R4	Digital Communications – Sanjay Sharma

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/117/105/117105143/
2	https://nptel.ac.in/courses/117101051/